

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: data = pd.read_csv('titanic_data.csv')
data
```

Out[2]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.25
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.28
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.92
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.10
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.05
...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.73

891 rows × 12 columns



In [3]:

data.shape

Out[3]: (891, 12)

```
In [4]: data.describe()
```

Out[4]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
In [5]: data.describe(include = 'object')
```

Out[5]:

	Name	Sex	Ticket	Cabin	Embarked
count	891	891	891	204	889
unique	891	2	681	147	3
top	Mangan, Miss. Mary	male	347082	G6	S
freq	1	577	7	4	644

```
In [6]: data.isnull().sum()
```

Out[6]:

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2
dtype:	int64

```
In [7]: data['Age'] = data['Age'].fillna(np.mean(data['Age']))
```

```
In [8]: data['Cabin'] = data['Cabin'].fillna(data['Cabin'].mode()[0])
```

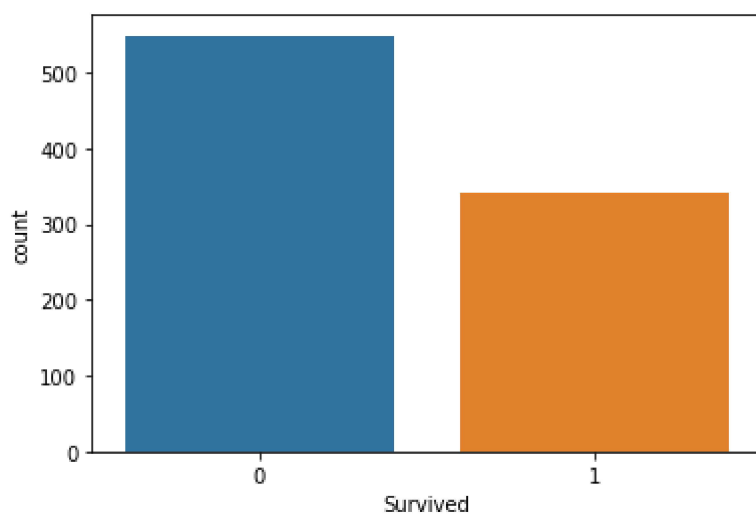
```
In [9]: data['Embarked'] = data['Embarked'].fillna(data['Embarked'].mode()[0])
```

```
In [10]: data.isnull().sum()
```

```
Out[10]: PassengerId    0
Survived              0
Pclass               0
Name                 0
Sex                  0
Age                  0
SibSp                0
Parch                0
Ticket               0
Fare                 0
Cabin                0
Embarked             0
dtype: int64
```

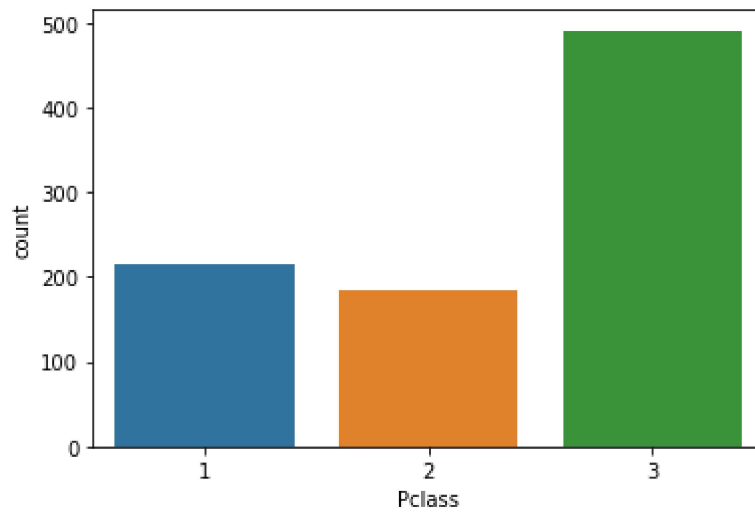
```
In [11]: sns.countplot(data['Survived'])
```

```
Out[11]: <matplotlib.axes._subplots.AxesSubplot at 0x1b5c823e7c0>
```



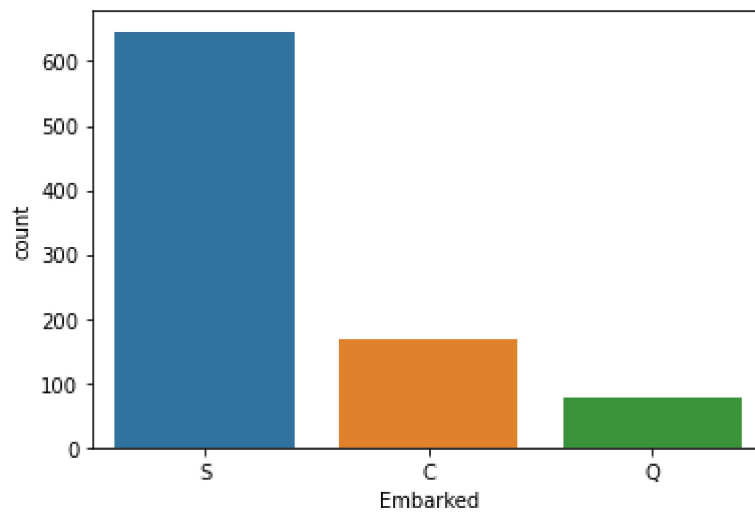
```
In [12]: sns.countplot(data['Pclass'])
```

```
Out[12]: <matplotlib.axes._subplots.AxesSubplot at 0x1b5c8997c70>
```



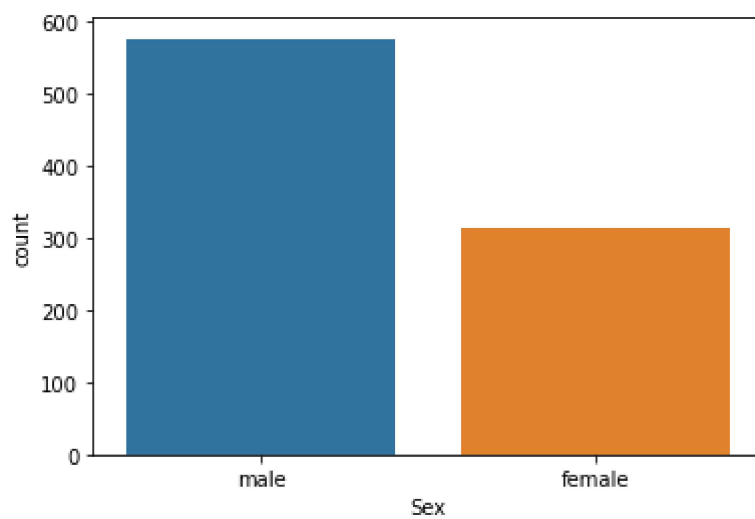
```
In [13]: sns.countplot(data['Embarked'])
```

```
Out[13]: <matplotlib.axes._subplots.AxesSubplot at 0x1b5c89f9910>
```



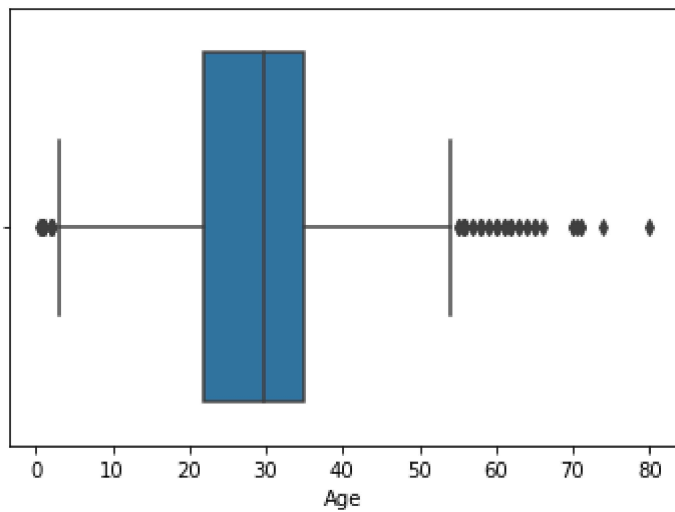
```
In [14]: sns.countplot(data['Sex'])
```

```
Out[14]: <matplotlib.axes._subplots.AxesSubplot at 0x1b5c8a4e8b0>
```



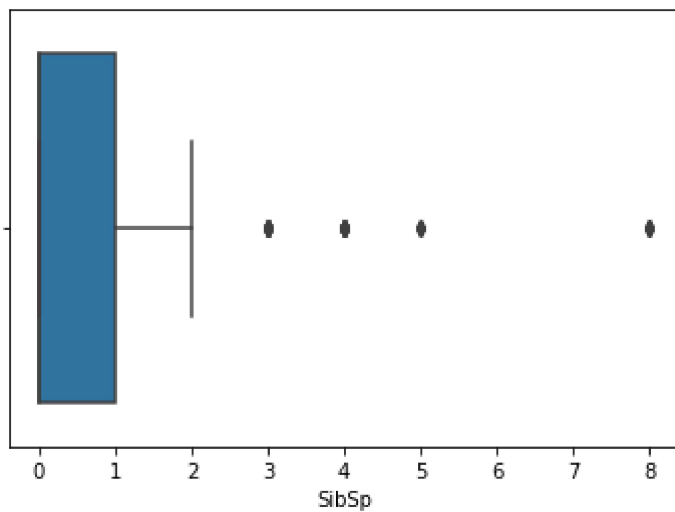
```
In [15]: sns.boxplot(data['Age'])
```

```
Out[15]: <matplotlib.axes._subplots.AxesSubplot at 0x1b5c8a9c490>
```



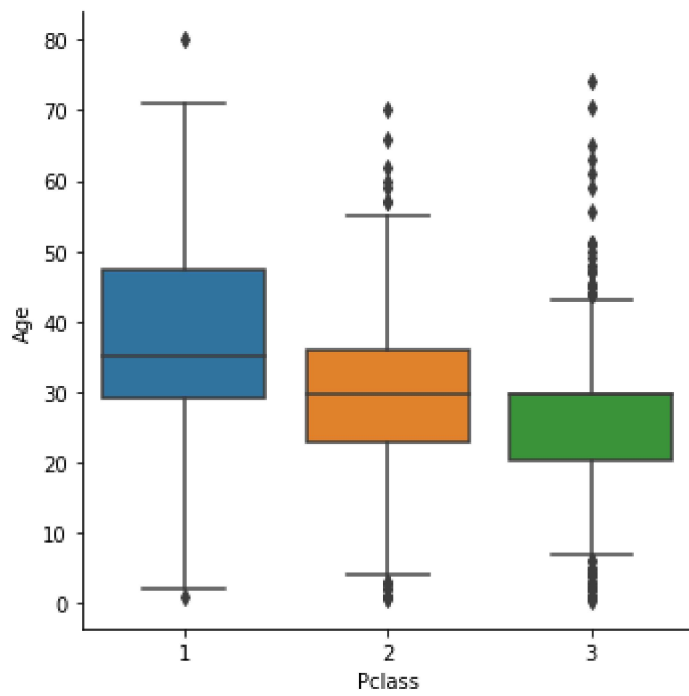
```
In [18]: sns.boxplot(data['SibSp'])
```

```
Out[18]: <matplotlib.axes._subplots.AxesSubplot at 0x1b5c8b9f940>
```



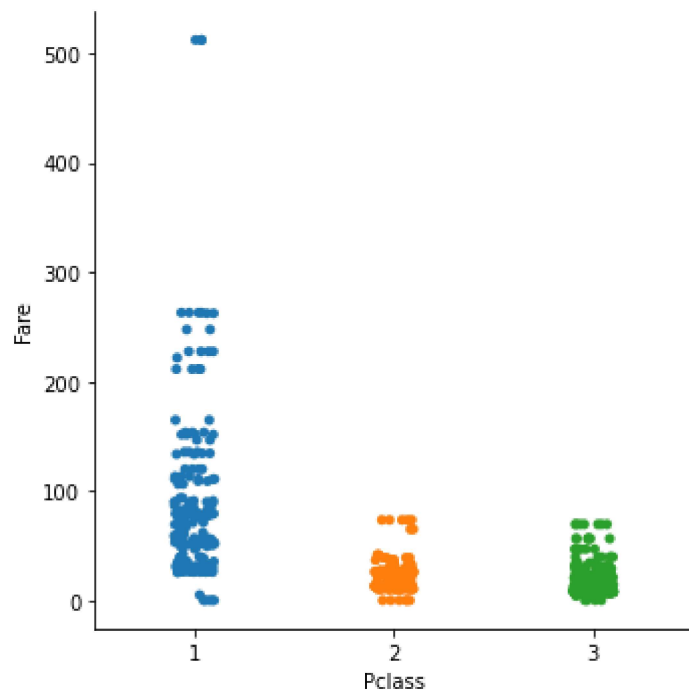
```
In [19]: sns.catplot(x= 'Pclass', y = 'Age', data=data, kind = 'box')
```

```
Out[19]: <seaborn.axisgrid.FacetGrid at 0x1b5c8affca0>
```



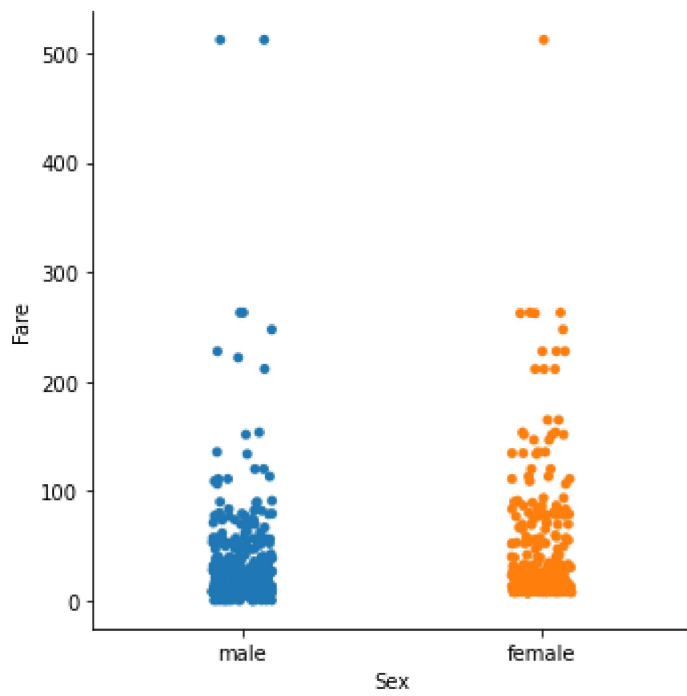
```
In [20]: sns.catplot(x= 'Pclass', y = 'Fare', data=data, kind = 'strip')
```

```
Out[20]: <seaborn.axisgrid.FacetGrid at 0x1b5c8ca3ee0>
```



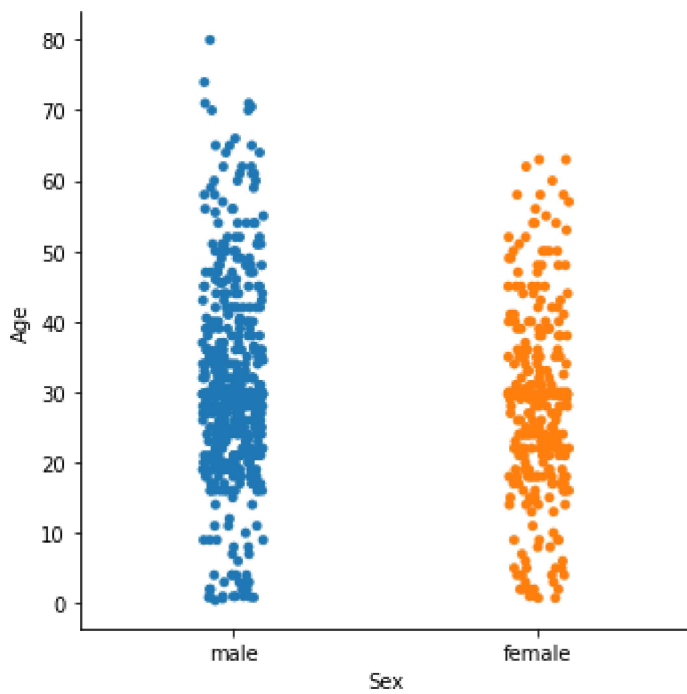
```
In [21]: sns.catplot(x= 'Sex', y = 'Fare', data=data, kind = 'strip')
```

```
Out[21]: <seaborn.axisgrid.FacetGrid at 0x1b5c8aed9a0>
```



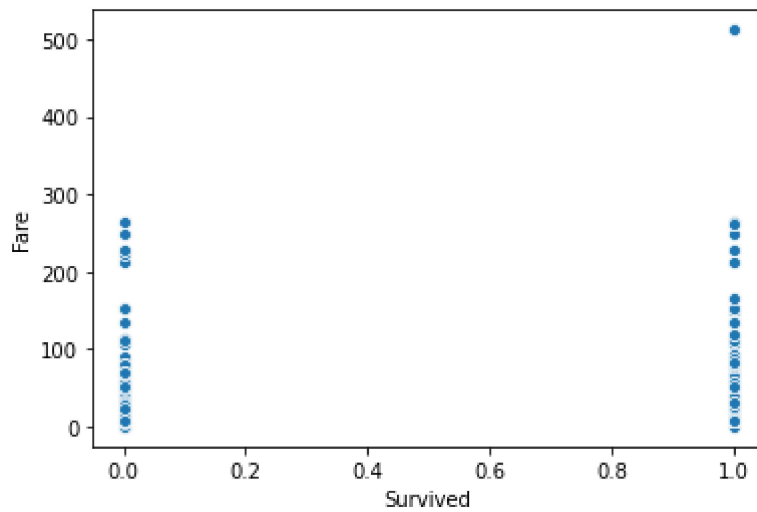
```
In [22]: sns.catplot(x= 'Sex', y = 'Age', data=data, kind = 'strip')
```

```
Out[22]: <seaborn.axisgrid.FacetGrid at 0x1b5c8d515e0>
```



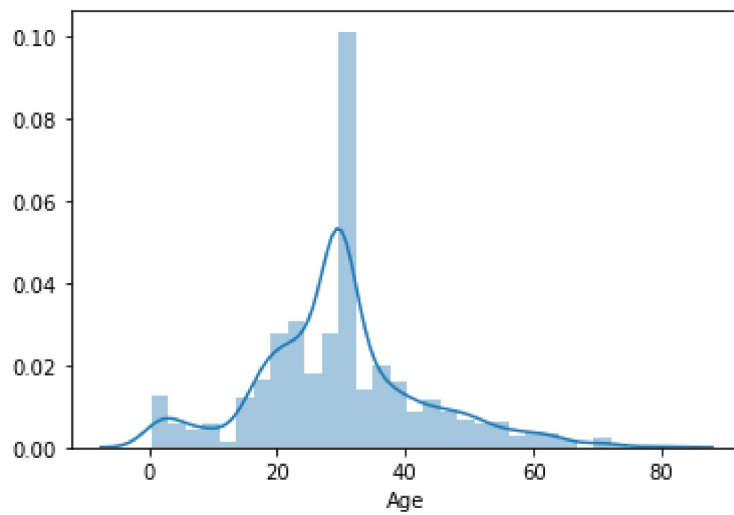
```
In [25]: sns.scatterplot(x = 'Survived', y = 'Fare', data = data)
```

```
Out[25]: <matplotlib.axes._subplots.AxesSubplot at 0x1b5c8d84820>
```



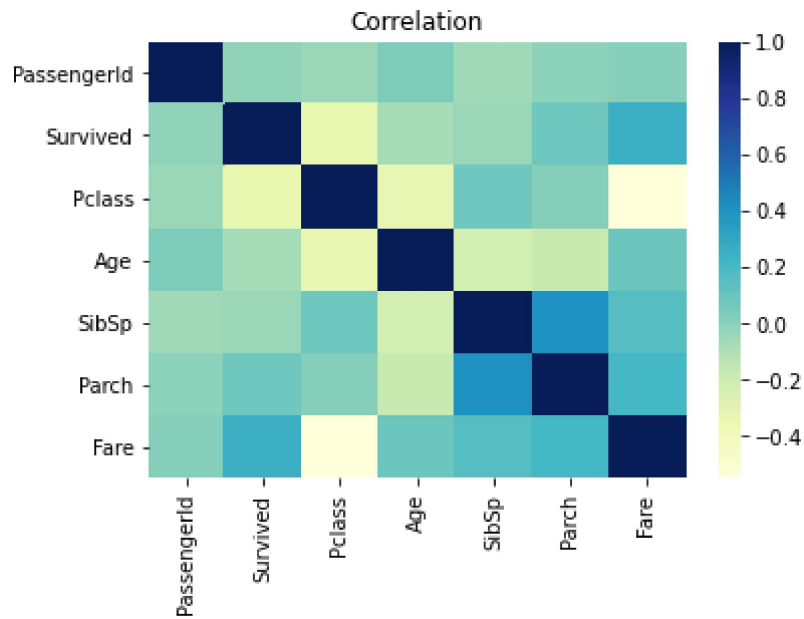
```
In [26]: sns.distplot(data['Age'])
```

```
Out[26]: <matplotlib.axes._subplots.AxesSubplot at 0x1b5cba89460>
```



```
In [29]: tc = data.corr()  
sns.heatmap(tc, cmap="YlGnBu")  
plt.title('Correlation')
```

```
Out[29]: Text(0.5, 1.0, 'Correlation')
```



Price of Ticket for each passenger is distributed

```
In [33]: sns.catplot(x='Pclass', y='Fare', data=data, kind='bar')
```

```
Out[33]: <seaborn.axisgrid.FacetGrid at 0x1b5ca67bcd0>
```

